

RESEARCH ARTICLE

Decoupling Volume and Magnetic Anisotropy for the Enhancement of Thermal Stability in CoFeB Multilayered Structures

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ABSTRACT

We demonstrate a fabrication strategy for multilayered CoFeB structures that overcomes the intrinsic trade-off between magnetic volume and perpendicular magnetic anisotropy, enabling a substantial enhancement in thermal stability. In conventional CoFeB/MgO heterostructures, interfacial anisotropy restricts the ferromagnetic layer thickness, severely limiting the magnetic volume and thermal robustness. This limitation is eliminated using a continuous vacuum process combining thin-film deposition, annealing, and an intermediate reverse-sputtering step to remove temporary MgO layers. The resulting multilayer stacks consist of ferromagnetically coupled CoFeB layers separated by thin Ta interlayers, yielding a significantly increased effective magnetic volume. Compared with a single-layer reference, the estimated thermal stability factor, calculated from the experimentally determined K_{eff} and magnetic volume, is enhanced by 400%. The multilayer structure is intrinsically asymmetric, allowing direct integration with adjacent spin-source layers for efficient net spin orbit torque-driven magnetization switching. These results establish a viable route toward thermally robust perpendicular magnetic devices suitable for next-generation spintronic applications.

Perpendicular magnetic tunnel junctions (MTJs) stand as the cornerstone technology for magnetic random-access memory, which promises high density, nonvolatility, and rapid switching speeds [1–3]. The fundamental magnetic and transport properties of these devices are largely governed by the CoFeB/MgO interface, which is critical for both high tunneling magnetoresistance (TMR) at room temperature and the requisite perpendicular magnetic anisotropy (PMA) for scalability [4–6]. However, the PMA in CoFeB/MgO is predominantly interfacial in nature [7], limiting the thickness of the CoFeB layer to approximately 1 nm to maintain the perpendicular easy axis.

This constraint forces a compromise between PMA and thermal stability (Δ), which is a key research imperative for long-term data retention at room temperature [8–10]. The thermal stability can be defined by $\Delta = K_{\text{eff}} \times V / (k_B \times T)$, where K_{eff} is the effective anisotropy energy, V is the magnetic volume, k_B is the

Boltzmann constant, and T is temperature [8, 11]. Achieving adequate thermal stability in CoFeB-based devices requires maximizing the $K_{\text{eff}} \times V$ product, which in turn demands either strong PMA or a sufficiently large magnetic volume. However, simply increasing the CoFeB thickness rapidly degrades the interfacial PMA, thereby offsetting any thermal stability gains from the increased volume. An alternative strategy is to circumvent interfacial anisotropy altogether by exploiting shape anisotropy, for example, by fabricating tall nanopillars that enforce PMA through geometrical confinement [12, 13]. However, such fabrication method is complex. Similarly, bulk PMA in rare earth-transition metal ferrimagnets, such as FeGd and FeTb, has been demonstrated to effectively decouple magnetic volume from interfacial constraints [14–16]. These systems have recently shown electrically driven switching in relatively thick magnetic films, highlighting their potential for thermally robust spintronic applications. However, CoFeB/MgO-based structures remain highly

attractive for MTJ technologies because of their strong interfacial PMA and high TMR. Another method to increase the magnetic volume is through multilayering of CoFeB, which has shown improvement in TMR [17, 18]. Dual-layer CoFeB stacks, MgO/CoFeB/Ta/CoFeB/MgO [7, 19–22], have been extensively employed to enhance V , this approach is not scalable beyond two CoFeB layers. Moreover, the oxide layers sandwiching the stack produce a symmetric structure that is inherently incompatible with spin-orbit torque (SOT) switching of the ferromagnetically coupled CoFeB layers.

Here, we introduce a scalable fabrication process that overcomes these limitations by enabling ferromagnetically coupled, multilayered CoFeB/Ta stacks with robust PMA. The process is implemented entirely under continuous vacuum and integrates physical vapor deposition, thermal annealing, and a critical reverse sputtering step. This approach allows the magnetic volume to be precisely scaled through the stacking of multiple CoFeB layers, thereby enhancing thermal stability while simultaneously yielding an asymmetric structure that is compatible with SOT operation.

The multilayered structure was fabricated using an 8-gun AJA magnetron sputtering system with a base pressure of 3×10^{-8} Torr on thermally oxidized Si substrates. All depositions were carried out within the growth chamber without breaking vacuum between layers, as illustrated in Figure 1. The CoFeB thickness was optimized to maximize interfacial PMA while maintaining robust magnetic properties. Based on the interfacial anisotropy model for CoFeB/MgO systems and experimental thickness-series optimization, a CoFeB thickness of 1.0 nm was selected, allowing the interfacial anisotropy to overcome the bulk and demagnetization contributions and establish a perpendicular easy axis. Metallic layers were deposited by DC sputtering at 50 W, while MgO layers were deposited by radio frequency (RF) sputtering at 100 W. MgO removal was performed under RF conditions at 50 W. Deposition pressures were maintained at 2 mTorr for all metallic layers and 1 mTorr for MgO growth, with an elevated working pressure of 20 mTorr employed during MgO removal to enhance etch uniformity. As annealing promotes the formation of an ordered CoFeB(001)/MgO(001) interface, which gives rise to strong interfacial PMA, a postdeposition vacuum anneal was performed at 300 °C for 1 h under PID temperature regulation, followed by a 4-h cooling cycle to ensure that the samples reached room temperature. The detailed

temperature ramp and cooling profiles are provided in Figure S1 of the Supporting Information. For additional magnetic layers, the cyclical process is initiated by etching away the MgO layer via reverse sputtering with argon ion bombardment. This step is immediately followed by the deposition of a thin Ta interlayer (0.5 nm). The Ta interlayer thickness was chosen to promote ferromagnetic coupling between adjacent CoFeB layers while simultaneously serving as a seed layer for subsequent CoFeB deposition. The cycle of deposition, annealing, and cooling is repeated to obtain the desired number of CoFeB layers. Finally, the stack is capped with a 4 nm-thick Ru layer for protection against ambient hydrolysis. Deposition and etch rates were calibrated independently by atomic force microscopy of film thickness after timed deposition and timed reverse-sputtering runs performed under identical process conditions. The optimization of the reverse-sputtering duration was critical to achieving PMA and the single-switching behavior observed in the ferromagnetically coupled structures. The reverse-sputtering duration was based on the calibrated rate and further refined with vibrating sample magnetometry (VSM) measurements to ensure that the reverse-sputtering process did not remove or damage the underlying CoFeB layer. The corresponding VSM results are shown in Figure S2a of the Supporting Information. When there is insufficient MgO removal, there are two distinct switching events, indicating that the CoFeB layers are not ferromagnetically coupled.

All fabrications were conducted using an automated sputtering system with preset recipes to ensure high reproducibility across the extended processing time.

Three structures were fabricated for comparative analysis, where the single CoFeB layer structure (CFB1) serves as the reference control for the double (CFB2) and triple (CFB3) structures. The details of the structure are shown as follows: CFB1: Ta(3)/CoFeB(1)/MgO(2)/Ru(4), CFB2: Ta(3)/CoFeB(1)/Ta(0.5)/CoFeB(1)/MgO(2)/Ru(4), and CFB3: Ta(3)/[CoFeB(1)/Ta(0.5)] $\times$ 2/CoFeB(1)/MgO(2)/Ru(4), where the number in parentheses represents thickness in nm.

The magnetic properties of the fabricated stacks were characterized using VSM with out-of-plane magnetic field sweeps. As shown in Figure 2, all stacks exhibit robust PMA with high remanence and a square-shaped hysteresis loop, demonstrating that the multilayer fabrication process successfully maintains the perpendicular easy axis. The slight reduction in magnetic

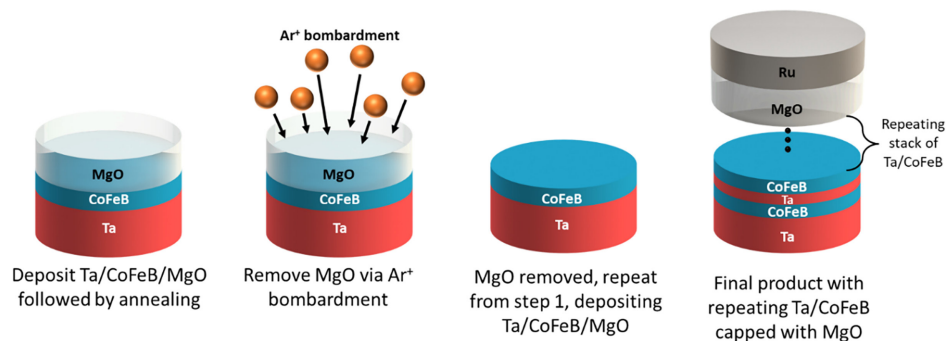


FIGURE 1 | Schematic of the multilayer fabrication sequence. A Ta/CoFeB/MgO trilayer is first deposited and annealed to promote interfacial crystallinity and PMA. The MgO layer is then removed via reverse sputtering, and the next Ta/CoFeB bilayer is deposited. This deposition-anneal-removal cycle is repeated to build the multilayer stack, which is finally capped with Ru. Layer thicknesses are not to scale.

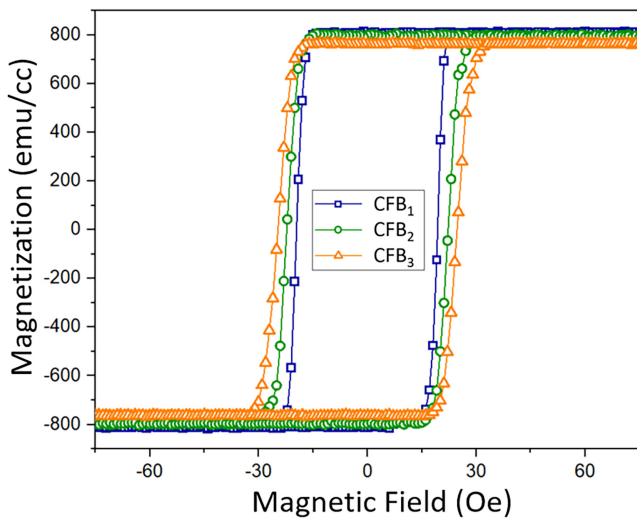


FIGURE 2 | Out-of-plane magnetic field sweeps measured by VSM for the CFB1, CFB2, and CFB3 multilayer stacks. The square-shaped hysteresis loops demonstrate ferromagnetically coupled CoFeB with robust PMA in all structures.

saturation with increasing layers is nominal and may be attributed to an increase in magnetic dead-layer from the additional CoFeB interfaces. Furthermore, the single sharp magnetic switching in every sample also suggests that the CoFeB layers are ferromagnetically coupled. Figure S2b in the Supporting Information provides the magnetostatic coupling strength as a function of spacer thickness for two CoFeB layers. Together with the VSM measurements in Figure S2a, these results indicate that ~ 1.1 nm of residual MgO thickness decouples the two CoFeB layers, resulting in dual switching. It should be noted that, in addition to reducing the coupling, residual MgO layers are also undesired as they increase the resistance-area product.

The anisotropy field (H_k) for each stack was quantified using Anomalous Hall effect (AHE) measurements performed on patterned Hall bars. For quantitative determination of H_k , AHE

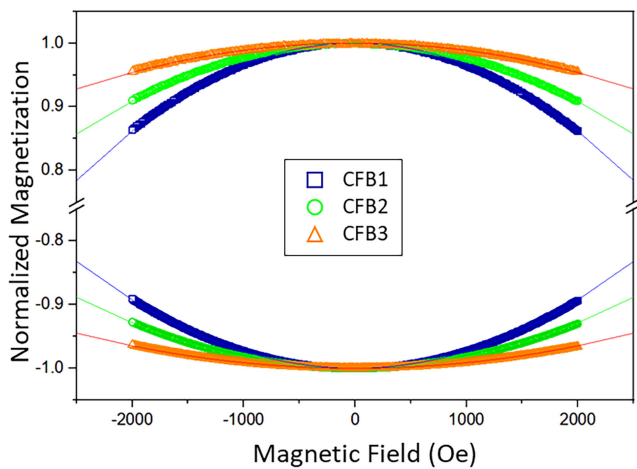


FIGURE 3 | Normalized anomalous Hall resistance, R_{xy} measured under a sweeping in-plane magnetic field for CFB1, CFB2, and CFB3. The solid curves represent the parabolic fits used to extract the anisotropy field H_k .

was preferred due to its superior signal-to-background ratio for ultrathin PMA films on Si substrates. As depicted in Figure 3, the Hall resistance R_{xy} was measured with an in-plane sweeping magnetic field. The anisotropy field, H_k was extracted from the normalized AHE measurement with a parabolic fitting given by $R_H^n = aH^2 \pm 1$ [23–25], where $\pm$ refers to the magnetization direction. The fitting range for a was from -2000 to $+2000$ Oe. Consequently, the H_k corresponds to the intersection of the fitted parabolic curve and the x axis, given by $H_k = \pm \sqrt{1/a}$. Details of the fitting parameters, fitting uncertainties, and extracted magnetic quantities are provided in Figure S3 of the Supporting Information.

The K_{eff} is subsequently derived from the measurement using the expression $K_{\text{eff}} = \frac{1}{2}H_k M_s$, where H_k was obtained from the AHE measurements and M_s was determined by VSM. Here, The enhancement in K_{eff} is primarily attributed to the additive interfacial anisotropy provided by the additional CoFeB/MgO interfacial layers and the ferromagnetic coupling between the CoFeB layers. Although the lower CoFeB layers experience longer cumulative thermal exposure during the cyclic fabrication process, which may introduce localized interfacial degradation or interdiffusion, the multilayer stacks continue to exhibit strong PMA and single-step magnetic switching. This cumulative effect aligns with established trends in multilayer PMA systems, where the net anisotropy enhancement originates from the collective contribution of multiple coupled magnetic interfaces [8, 26–28].

Hence, given that the magnetic volume has approximately doubled and tripled with CFB2 and CFB3, respectively, the thermal stability factor is determined to be 240% and 400% higher than the control CFB1. Figure 4 illustrates the percentage increments for coercivity H_c , K_{eff} , and thermal stability Δ , demonstrating that the main driver of the pronounced increase in the thermal stability is mostly derived from the larger magnetic volume rather than from the subtle improvements in K_{eff} . While increasing the effective magnetic volume enhances the thermal stability factor, it is also expected to increase the critical switching current in both STT- and SOT-based MRAM devices. The proposed architecture therefore represents a trade-off between

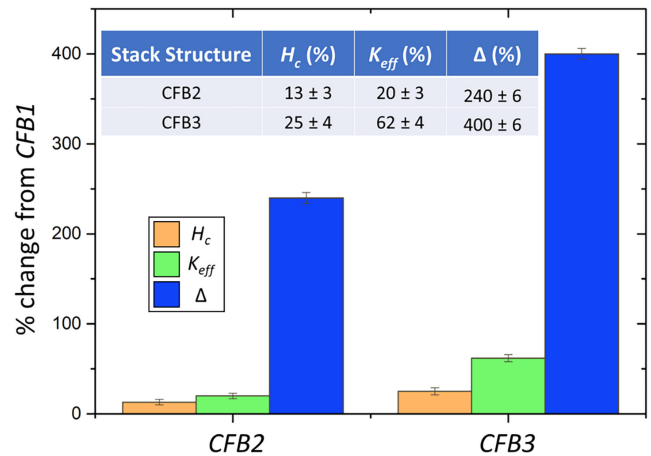


FIGURE 4 | Coercivity H_c , K_{eff} , and Δ as percentage improvement of CFB2 and CFB3 with CFB1 as the benchmark.

retention and switching energy. By controlling the number of ferromagnetically coupled CoFeB layers, the energy barrier can be tailored to meet specific application requirements, particularly in scenarios where enhanced data retention and thermal robustness are prioritized.

To further verify the removal of the MgO layer during the reverse sputtering process, cross-sectional transmission electron microscopy (TEM) and energy-dispersive X-ray (EDX) spectroscopy were performed on CFB3. CFB3 was selected as the representative sample for TEM and EDX characterization because it underwent the largest number of deposition, annealing, and reverse-sputtering cycles and is therefore the most susceptible to residual MgO accumulation or interface degradation. Structural validation of CFB3 thus serves as a stringent verification of the fabrication process. The TEM micrograph shown in Figure 5a reveals a single white line indicative of the 2 nm MgO layer below the Ru capping layer. This verifies that the reverse sputtering has effectively removed the intermediate 2 nm MgO layers prior to each subsequent deposition cycle, validating the key step of our fabrication method. The layer thicknesses observed in Figure 5a may appear slightly larger than their nominal values due to projection effects, specimen tilt, and interfacial contrast broadening inherent to cross-sectional TEM imaging. The nominal thicknesses reported in this work are based on independently calibrated deposition and etch rates. While the ultrathin 0.5 nm Ta interlayers are not individually resolved, the TEM and EDX analyses confirm the preservation of the multilayer structure and the successful removal of the intermediate MgO layers following repeated processing cycles. The corresponding EDX line scan in Figure 5b shows that the elemental peaks are closely spaced due to the ultrathin nature of the film stack. However, the layering of the Ru cap, followed by MgO, Fe, and Ta, can be observed. The majority of the Ta signal is localized to the 3 nm Ta seed layer, and the 0.5 nm interlayers between the CoFeB layers did not contribute a resolvable signal, consistent with their subnanometre thickness.

The demonstrated structure also presents a key advantage for SOT integration with MTJs. Unlike conventional dual CoFeB-based structure, in which the SOT contribution from the sandwiched spin-source layer is minimized due to the opposing torques in the ferromagnetically coupled CoFeB layers. In contrast, the proposed

asymmetric multilayer structure in this work allows for the entire ferromagnetic volume to be adjacent to a single spin-source layer, enabling a net torque for efficient current-driven magnetization switching. Figure 6 conceptually demonstrates the difference, where the highlighted spins on the right of the figure represent the net torque acting on the ferromagnetically coupled CoFeB layers in the proposed structure, in contrast to the compromised net torque in the conventional stack on the left. The enlarged free layer in the figures serves to facilitate the illustration of the spin-polarized electron transport within the multilayer structure and should not be taken as a structural requirement. While the integrated structure described here provides a potential pathway toward enhanced net SOT efficiency, the focus of this work remains the architectural validation and the substantial gain in thermal stability. Figure 6 schematically illustrates a potential advantage of the proposed asymmetric architecture for future SOT devices. However, no direct SOT measurements are presented in this work, and therefore, the discussion is intended as a conceptual outlook rather than an experimentally validated result.

In summary, we have developed and experimentally validated a cyclic reverse-sputtering fabrication strategy for multilayered CoFeB structures that enhances thermal stability while maintaining strong PMA. By sequentially establishing PMA in individual CoFeB layers and coupling them through ultrathin Ta interlayers, the architecture increases the effective magnetic volume beyond that of a single continuous CoFeB layer. Magnetic characterization shows an approximately fourfold enhancement

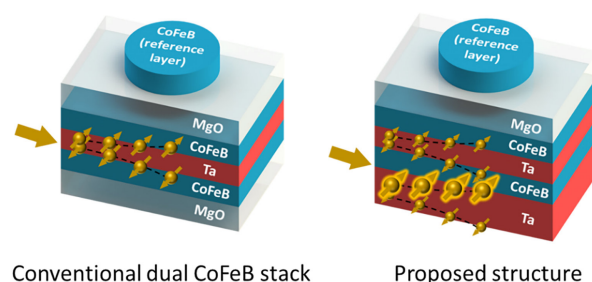


FIGURE 6 | Illustration of the spin-orbit torque enablement in the proposed asymmetric multilayer stack compared with a conventional stack.

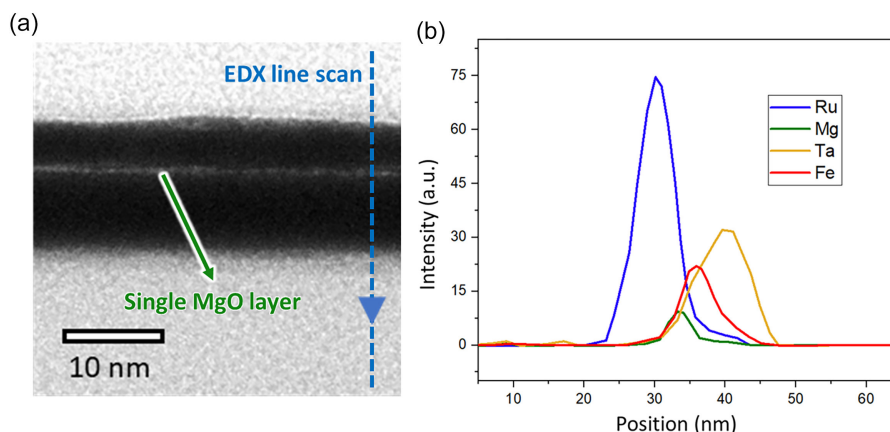


FIGURE 5 | (a) Cross-sectional TEM micrograph of the CFB3 structure. (b) EDX line scan displaying the elemental intensity profiles for Ru, Mg, Ta, and Fe across the film thickness.

in the estimated thermal stability factor for the three-layer CoFeB structure compared with the single-layer control while preserving robust PMA and single-step magnetization switching. Cross-sectional TEM and EDX analyses confirm the effective removal of intermediate MgO layers and the structural integrity of the multilayer stack after repeated processing cycles. This approach provides a scalable route to increase magnetic volume while retaining the interfacial anisotropy required for CoFeB/MgO-based devices, offering a promising pathway toward thermally robust perpendicular MTJs and high-density spintronic memory technologies.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. See Supplementary Information for details on temperature profile for the annealing process, VSM measurements with varying MgO removal duration with simulated magnetostatic coupling strength and details on magnetics parameters.